Name: ________________________ ID: ________________________

Discrete Mathematics SC205 (Winter 2023-2024)
First In-semester Examination
DA-IICT, Gandhinagar, Gujarat, India

1, 2, 8, 5

Date: 07/02/2024 Total Marks: 30 Duration: 90 Mins.

Instructions: Write your name and ID on all pages failing so your answer sheet will not be evaluated. All questions are compulsory. Write your answer immediately next to the question on the same page. Additional sheets will not be evaluated. Answers must have **proper justifications** to get full credit.

1. [2 marks] Use a truth table to verify whether the two expressions are equivalent.
$\neg(p \vee (q \wedge (\neg r)))$ and $(\neg p) \wedge ((\neg q) \vee r)$.

p	q	r	$q \wedge (\neg r)$	$\neg(p \vee (q \wedge (\neg r)))$	$(\neg q) \vee r$	$(\neg p) \wedge ((\neg q) \vee r)$
1	1	1	0	0	1	0
1	1	0	1	0	0	0
1	0	1	0	0	1	0
0	1	1	0	1	1	1
0	0	0	0	1	1	1
0	0	1	0	1	1	1
0	1	0	1	0	0	0
1	0	0	0	0	1	0

Equivalent

✓

If correct then full marks

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2. **[4 marks]** Let the proposition $G(x)$ stand for "x believes in God", $S(x)$ stand for "x is a scientist", $P(x, y)$ stand for "x is a parent of y, where the universe of discourse U is the set of all living people. For each of the following propositions, **state and justify** if it is True or False:

(a) $(\exists x \text{ in } U)(S(x) \wedge G(x))$

(b) $(\exists x \text{ in } U)(\neg S(x) \wedge \neg G(x))$

(c) $(\forall x \text{ in } U)(\exists y \text{ in } U)(P(x, y))$

(d) $(\exists x \text{ in } U)(\exists y \text{ in } U)(P(x, y) \wedge \neg G(x) \wedge G(y))$

a) True, there exist a person who is scientist and believe in god.

b) True, there exist a person who is not scientist and not believe in god

c) Not True, for every person in U, there exist atleast one person who is their parent (There exist person who are not parent)

d) There exist x and y such that x is a parent of y and x not belives in God and y belives in god. True

- If correct full marks
- If ~~Ea~~ correct answer but with no justification the 1/2 marks
- If justification is wrong and answer is correct then 1/2 marks

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3. [4 marks] Prove that for a positive integer n, n is even if and only if $7n+4$ is even.

Ans:

Let 'n' be a positive integer.
We have to prove that 'n' is even if and only if $7n+4$ is even.

(i) Let 'n' is even

To prove: $7n+4$ is even

Since 'n' is even; so $n = 2k_1$, $k \in \mathbb{Z}$

So $7n+4 = 7(2k_1)+4$

$= 14k_1+4$

$= 2(7k_1+2)$

So $7n+4 = 2m$, $m = 7k_1+2 \in \mathbb{Z}$

Hence $7n+4$ is even.

(ii) Let $7n+4$ is even.

So $7n+4 = 2k_2$; $k_2 \in \mathbb{Z}$

$\Rightarrow 7n = 2k_2 - 4$

$\Rightarrow n = \dfrac{2(k_2-2)}{7}$

Hence $\dfrac{k_2-2}{7} = p \in \mathbb{Z}$, if this is not an integer; then "k_2" will not be an integer.

So $n = 2p$

Hence 'n' is even. ($\square$)

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4. [4 marks] Let $f : \mathbb{R} \to \mathbb{Z}$ be a function from the set of real numbers to the set of integers and is defined as $f(x) = \lfloor \frac{x+1}{4} \rfloor$. Prove whether f is one-to-one and onto.

Ans.

Let $f : \mathbb{R} \to \mathbb{Z}$ be a function from set of real numbers to the set of integers defined by

$$f(x) = \left\lfloor \frac{x+1}{4} \right\rfloor$$

The function is not one-one, because

$$f(0) = \left\lfloor \frac{0+1}{4} \right\rfloor = \left\lfloor \frac{1}{4} \right\rfloor = 0$$

$$f(1) = \left\lfloor \frac{1+1}{4} \right\rfloor = \left\lfloor \frac{2}{4} \right\rfloor = \left\lfloor \frac{1}{2} \right\rfloor = 0$$

Since $0 \neq 1$ but $f(0) = f(1)$ so function is not one-to-one.

The function is onto, because let $y \in \mathbb{Z}$;

so $4y \in \mathbb{Z}$

$\Rightarrow 4y - 1 \in \mathbb{Z}$

so given any $y \in \mathbb{Z}$; we can have $4y-1 \in \mathbb{Z}$

such that

$$f(x) = f(4y-1) = \left\lfloor \frac{4y-1+1}{4} \right\rfloor$$

$$= \left\lfloor \frac{4y}{4} \right\rfloor$$

$$= \lfloor y \rfloor$$

$$= y \quad (\text{as } y \in \mathbb{Z})$$

So function is not one-to-one, but onto.

$(\square)$

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5. [4 marks] Prove or give a counterexample for the following for three non-trivial non-empty sets: A, B, and C. Justify your answer.

(a) If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

(b) If $A \in B$ and $B \in C$, then $A \in C$.

a) Let x be arbitrary and $x \in A$

$A \subseteq B \Rightarrow x \in B$

$B \subseteq C \Rightarrow x \in C$

and x is arbitrary

$\Rightarrow A \subseteq C$.

b) $A = \{1,2\}$ $\quad B = \{\{1,2\}, 3, 4\}$, $C = \{\{\{1,2\},3,4\}, 5\}$

Clearly $A \in B$ and $B \in C$

But $A \notin C$.

a) • If mathematically proof then full marks

• Prove by example 0 marks.

b) • Counterexample then full marks

• $A = 1 \longrightarrow$ not a set X

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6. [4 marks] Define the relations R over the set of integers $\mathbb{Z}$ as follows: $a, b \in \mathbb{Z}$, $(a, b) \in R$ if and only if $a + 3b$ is divisible by 4. Prove whether R is an equivalence relation or not. (x is divisible by y if there exists an integer k, such that $x = k \times y$.)

Ans.

The relation 'R' is defined by (over the set of integers $\mathbb{Z}$) as follows:

$a, b \in \mathbb{Z}$, $(a,b) \in R$ iff $a+3b$ is divisible by 4

For proving Equivalence relation we have to prove that the relation is reflexive, symmetric and transitive.

Reflexive:-

$\forall a \in \mathbb{Z}$, $(a,a) \in R$, i.e.

So $a + 3a = 4a$ is divisible by 4

∴ The relation is reflexive.

Symmetric:-

$\forall a, b \in \mathbb{Z}$; if aRb then bRa

i.e. if $(a,b) \in R$ then $(b,a) \in R$

Since $(a,b) \in R$. So $a+3b = 4k_1 \Rightarrow a = 4k_1 - 3b$ — ①

Now $b+3a = b + 3(4k_1 - 3b) = b + 12k_1 - 9b$

$= 4(3k_1 - 2b)$ ∴

$3k_1 - 2b \in \mathbb{Z}$

So $(b,a) \in R$

Transitive:-

$\forall a,b,c \in \mathbb{Z}$, if aRb, and bRc then aRc

i.e. if $(a,b) \in R$ & $(b,c) \in R$ then $(a,c) \in R$.

Since $a+3b = 4k_1$ and $b+3c = 4k_2$, $k_1, k_2 \in \mathbb{Z}$

So $a + 3b + b + 3c = 4(k_1 + k_2)$

$\Rightarrow a + 3c = 4(k_1 + k_2 - b)$; $k_1 + k_2 - b \in \mathbb{Z}$

So it is an Equivalence relation.

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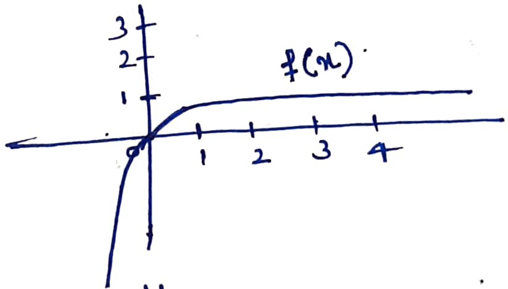
7. [4 marks] Let A be the set $A = \{x \in \mathbb{R} | x > 0\}$ and define $f, g, h : A \to \mathbb{R}$ by $f(x) = \frac{x}{1+x}$, $g(x) = \frac{1}{x}$, $h(x) = -x + 1$.
Find $g \circ f$, $f \circ g$, $g \circ h$, $h \circ g$.

Ans. $f : A \to \mathbb{R}$ is defined by

$f(x) = \frac{x}{1+x}$

Domain(f) = A

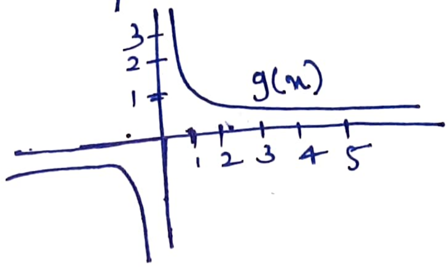
Range(f) = $(0,1)$



$g(x) = \frac{1}{x}$

Domain(g) = A

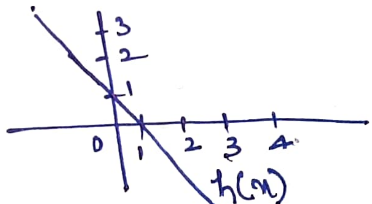
Range(g) = $(0,\infty)$



$h(x) = -x+1$

Domain(h) = A

Range(h) = $(-\infty,1)$



Hence $g \circ f : A \to \mathbb{R}$ is possible because Range$(f) \subseteq$ Domain(g)
and defined by $g \circ f = g\left(\frac{x}{1+x}\right) = 1 + \frac{1}{x}$

Similarly $f \circ g : A \to \mathbb{R}$; since Range$(g) \subseteq$ Domain(f)

$f \circ g = f\left(\frac{1}{x}\right) = \frac{1/x}{1+1/x} = \frac{1/x}{\frac{x+1}{x}} = \frac{1}{1+x}$

but; $g \circ h : A \to \mathbb{R}$ is not possible because
Range$(h) = (-\infty,1) \not\subseteq$ Domain$(g) = A = \{x \in \mathbb{R} | x > 0\}$

Also $h \circ g : A \to \mathbb{R}$ is possible ; since Range$(g) \subseteq$ Domain(h)

$h \circ g = h\left(\frac{1}{x}\right) = 1 - \frac{1}{x}$

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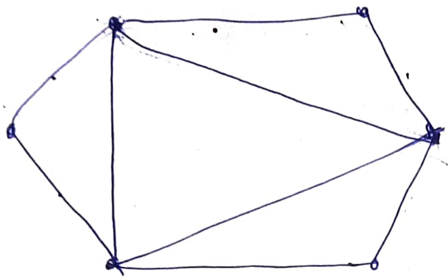
8. [4 marks] Answer the following questions.

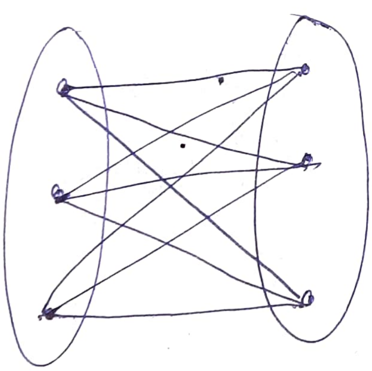
- Draw a simple graph with exactly 6 vertices and exactly 9 edges that is not bipartite but have an Euler circuit.
- Draw a simple graph with exactly 6 vertices and exactly 9 edges that is bipartite but do not have an Euler circuit.

a)



All vertices are even degree and not bipartite (as odd cycle is there

b)



6 vertices
9 edges
Bi~~bacr~~partite but not Euler as degree is odd

(a,b) - Only graph without any explanation then half marks
- Write proper that Euler circuit do not exist if any vertex has odd degree.
- Show Euler and bipartite also